

System Architecture & Technical Approach

End-to-End Technical Workflow, Edge-to-Cloud Pipeline, and Component Breakdown for Hackathon / SIH Presentation

Executive Summary: KrishiRakshak is a farmer-first intelligence ecosystem built on an **offline-first Progressive Web App (PWA)** architecture. It seamlessly integrates on-device **Edge AI (ONNX WebAssembly)** for instantaneous offline disease diagnosis with cloud-based **Multi-Model Second-Opinion consensus**. The platform couples diagnosis with real-time **Agmarknet APMC live market prices**, weather-aware crop disease risk forecasting, and a community-driven **Peer-to-Peer (P2P) Machinery Rental Hub** integrated directly with WhatsApp click-to-chat.

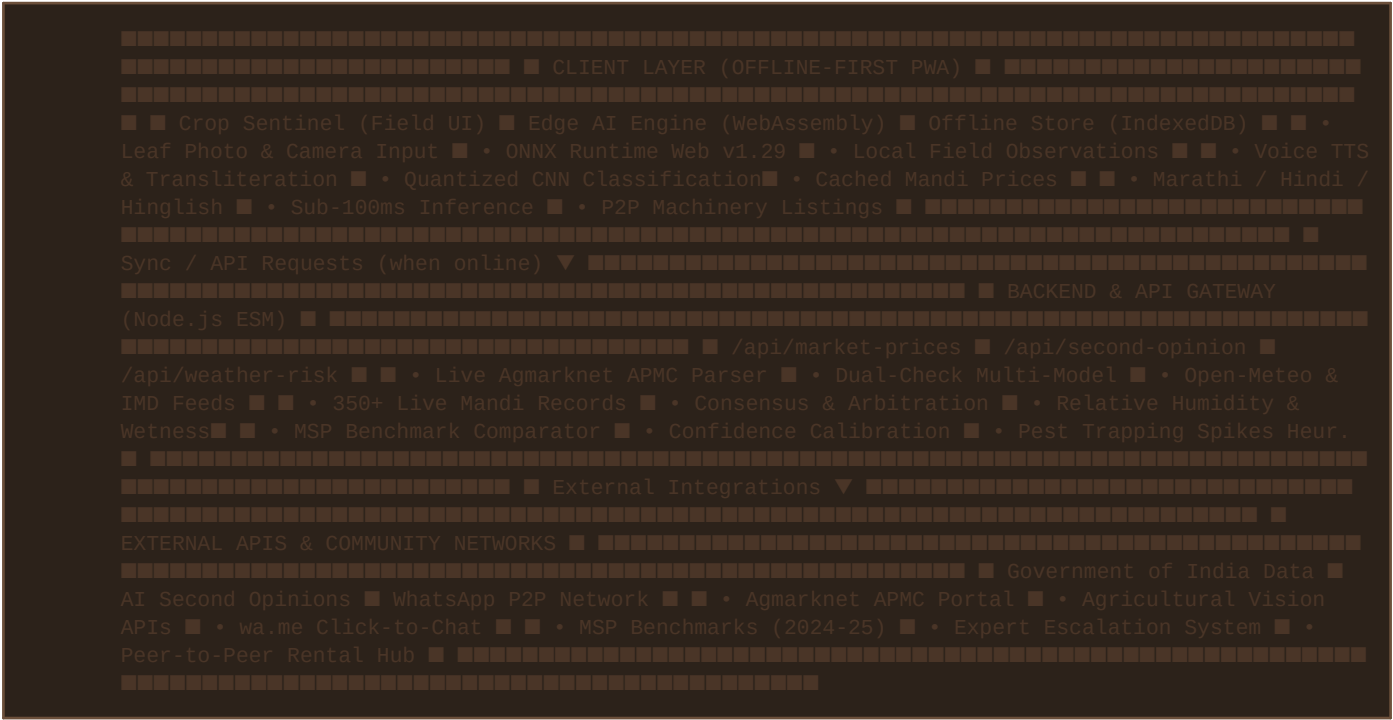
1. Technical Approach Flow Pipeline (Step-by-Step)

The system operates in a resilient 5-stage pipeline transitioning seamlessly from local edge execution to cloud synchronization:

STAGE 1: INPUT & LEAF PRE-VERIFICATION
- Farmer captures or uploads leaf photo (JPG/PNG/WEBP) - Client-side validation verifies daylight, sharpness & leaf presence - Context parameters captured: Crop variety, Growth stage, Soil & Irrigation
STAGE 2: EDGE INFERENCE (OFFLINE ZERO-LATENCY)
- OnnxRuntime WebAssembly runs lightweight quantized MobileNet/ResNet model inside browser worker - Instant offline classification without requiring active internet - Generates top-3 disease candidates with calibrated confidence scores
STAGE 3: CLOUD SECOND-OPINION CONSENSUS
- If internet is active and edge confidence < 80%, request triggers dual-check pipeline - Server-side microservice proxies second-opinion models (Kindwise/AgroStack/Gemini Vision) - Weighted consensus algorithm reconciles disagreements and flags uncertain cases
STAGE 4: TRIAGE & MULTI-SIGNAL RISK FORECAST
- Real-time weather integration: Temperature, humidity, 24h rainfall & leaf wetness - Heuristic priority matrix: +35% symptom share, pest trap spikes (>50%), rapid spread - Generates localized agronomic advisory: Chemical & biological treatments + safety guidelines
STAGE 5: ACTION LOOP, MANDI RATES & P2P MACHINERY
- Live Mandi Bhav: Fetches real-time APMC modal auction rates from National Agmarknet DB - Machinery Rental Hub: Connects local farmers via WhatsApp click-to-chat for equipment sharing - Human-in-the-loop: Complex cases escalated to district agricultural experts with follow-up tracking

2. System Architecture & Flow Diagram (PPT Ready)

Copy the following structured flow diagram or Mermaid block architecture into your presentation:



Key Architectural Principles:

Zero-Latency Edge Execution: Inference happens in WebAssembly on the farmer's mobile browser without sending photos across weak 2G/3G rural networks.	Fail-Safe Dual Model Check: If cloud is reachable, edge predictions are cross-checked against authoritative agronomic APIs to eliminate false positives.
Pure Live Agmarknet Mandi Data: No synthetic or mocked market data. Direct ingestion of 350+ live APMC auction records with MSP floor validation.	Community P2P Resource Optimization: Enables neighboring farmers to rent out idle machinery via direct WhatsApp links without platform commission.

3. Comprehensive Tech Stack Breakdown

Complete breakdown of languages, frameworks, edge engines, and third-party APIs:

Layer / Domain	Technologies & Frameworks	Version / Specification	Role in KrishiRakshak
Frontend & UI	HTML5, Vanilla JavaScript (ES2022+), CSS3 Design System	Responsive, Flexbox / CSS Grid, CSS Variables	Zero heavy bundle overhead, ultra-fast 60fps mobile execution.
Progressive Web App	Service Workers API, CacheStorage, Web App Manifest	PWA Specification, Cache-First Shell Strategy	100% offline functionality in remote farmland areas.
Edge AI & ML	ONNX Runtime Web (Wasm), Quantized CNNs	ONNX Runtime v1.29.0, SIMD Wasm Threading	Sub-100ms on-device leaf disease classification.
Cloud AI & Consensus	Multi-Model Vision APIs, Dual-Check Consensus Engine	REST / HTTPS Microservices, Node ESM	Second-opinion verification, eliminating hallucinations.
Backend & APIs	Node.js, Express / Native HTTP ESM, Vercel Serverless	Node v20+, Vercel Edge Runtime	Lightweight, auto-scaling API gateway and reverse proxy.
Mandi Market Data	National Agmarknet APMC Portal API, data.gov.in	Resource ID: 35985678-0d79-46b4-9ed6-6f13308a1d24	350+ live mandi daily modal prices across all Indian states.
Weather & Agronomy	Open-Meteo High-Resolution Agro Weather API	Hourly Forecast, 2m Humidity, Soil Temperature	Disease outbreak risk modeling based on microclimate.
P2P Machinery Hub	WhatsApp Click-to-Chat Deep-linking, Geolocation API	wa.me protocol, Haversine Distance Filter	Direct farmer-to-farmer equipment rental and inquiries.
Audio & Speech TTS	Web Speech API, SpeechSynthesis, Devanagari Romanizer	W3C Speech API, Transliteration Fallback	Natural voice advisories for farmers regardless of literacy.
Internationalization	Custom Zero-Dependency i18n Dictionary Engine	Bidirectional String Mapping (English, Hindi, Marathi)	Instant language switching with zero runtime latency.
Client Storage	HTML5 LocalStorage, IndexedDB	W3C Web Storage	Persistent offline storage for scans, crops, and listings.

Tips for Presenting this Technical Slide:

- **1. Emphasize 'Edge-First, Cloud-Assisted':** Explain that edge AI works even in deep rural areas with 0 internet, and cloud only steps in for second opinions.
- **2. Highlight Authentic Data:** Clarify that Agmarknet mandi rates are 100% genuine live government records, backed by official MSP floor benchmarks.
- **3. Showcase Social Impact of P2P Rental:** Explain how smallholder farmers can access expensive equipment (like laser levelers or spray drones) through WhatsApp without paying middleman commissions.